Unit 2 Cluster Practice: Systems of Matter and Energy

Earth & Space Science

Name: _____ Period: _____ Date: _____

Skill: Sequence the steps of a process.
Strategy: Decide which step has to happen first. Then write the letters in order and reread them as a story to check that each step leads to the next.

Base your answer to question 1 on the information below and on your knowledge of Earth and Space Sciences. Some questions may require the use of the *Reference Tables*.

The Changing Landscape of Niagara Falls

Niagara Falls formed about 12,000 years ago. The melting ice from retreating glaciers released fresh water that formed the Great Lakes. Water flowing from Lake Erie to Lake Ontario created the Niagara River. Niagara Falls separates the upper Niagara River from the lower Niagara River. The depth of the lakes changes throughout the year. The highest lake levels occur in spring and early summer.

1 Place the letters of the processes, from the choices below, in the correct sequence to describe the formation of sandstone in the Niagara escarpment. [1]

Choices:

A	B	C	D
Deposition of particles	Compaction of particles into layers	Weathering of land formations	Transport of particles driven by gravity

Process of Sandstone Formation

Letter: ____ → Letter: ____ → Letter: ____ → Letter: ____

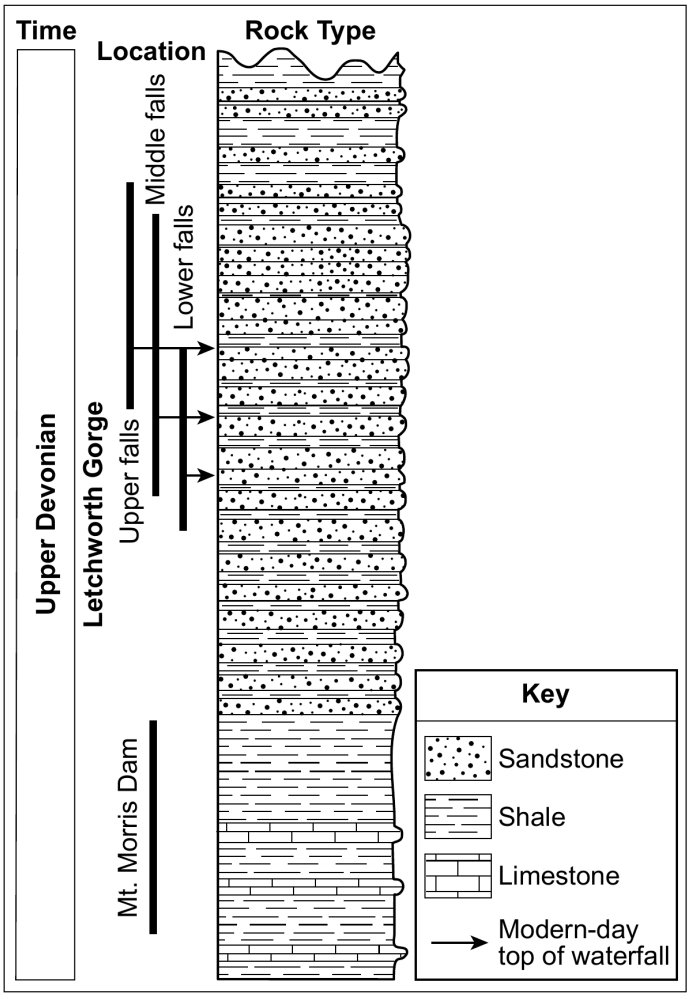
Skill: Use evidence from a model to identify processes.
Strategy: Look at the key. Decide what type of rock the layers are (igneous, sedimentary or metamorphic) first, then pick the processes that form that type of rock.

Base your answer to question 2 on the information below and on your knowledge of Earth and Space Sciences. Some questions may require the use of the *Reference Tables*.

Mount Morris Dam in Letchworth State Park

Letchworth Gorge is located southwest of Rochester, New York, where the Genesee River flows through the local landscape. It is sometimes referred to as the "Grand Canyon of the East" due to the height of its vertical face. The rock seen in the walls of the gorge consists primarily of shales and sandstones. These rocks formed during the Late Devonian Period. The Mount Morris Dam is located at the northern end of Letchworth Gorge.

Model of Rock Types for a Portion of Letchworth Gorge



(Not drawn to scale)

2 What are the constructive processes most likely responsible for the distribution of rock layers found in the Letchworth Gorge?

(1) recrystallization, deposition, and weathering of rocks

(2) solidification, cooling, and outgassing of sediments

(3) erosion, mineral precipitation, and folding of rocks

(4) burial, compaction, and mineral precipitation of sediments

Skill: Use observations as evidence to classify a rock.
Strategy: Underline each clue in the reading. Check every column in a row. If one part of a row is wrong, the whole row is wrong.

Base your answer to question 3 on the information below and on your knowledge of Earth and Space Sciences. Some questions may require the use of the *Reference Tables*.

Rocks of the Great Smoky Mountains

Great Smoky Mountains National Park sits along the border of Tennessee and North Carolina. Many of the rocks in the park are part of a rock unit that geologists call the Ocoee Supergroup.

A student collected a sample of one of these rocks. The rock is fine-grained, has no visible fossils, and splits into thin, flat sheets. Geologists found that the rock formed from shale that was later buried deep within Earth's crust when the Appalachian Mountains formed.

3 Which row in the table below correctly identifies the rock name, the rock-forming process, and the location of formation for the student's sample?

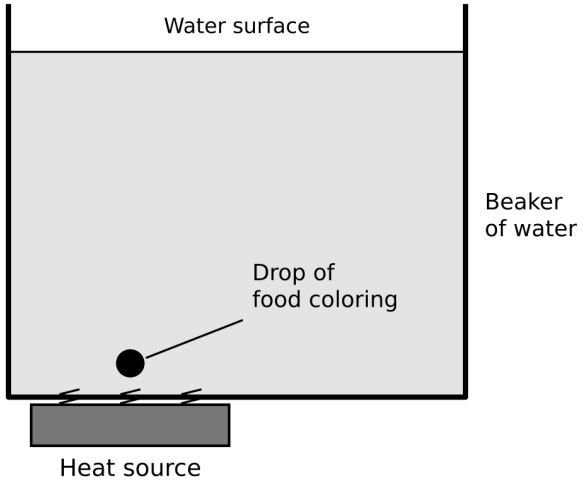
Row	Rock Name	Rock-Forming Process	Location of Formation
(1)	Slate	Heat and pressure	Deep within the crust during mountain building
(2)	Slate	Compaction and cementation	On an ocean floor
(3)	Shale	Heat and pressure	At Earth's surface
(4)	Gneiss	Melting and solidification	Deep within the crust during mountain building

Skill: Develop a model.
Strategy: Hot water is less dense, so it rises. Start your first arrow at the food coloring and follow the water all the way around. Then explain the cause using the word *density*.

Base your answer to question 4 on the information below and on your knowledge of Earth and Space Sciences. Some questions may require the use of the *Reference Tables*.

Modeling Convection

A student placed a beaker of room-temperature water so that only the left side of the beaker sat over a heat source. The student then placed a drop of food coloring on the bottom of the beaker, directly above the heat source. The diagram below shows the setup.



(Not drawn to scale)

4 On the diagram above, draw at least **two** arrows to show the path the food coloring will follow as convection occurs in the water. Then explain why the water directly above the heat source rises. [1]

Skill: Use evidence to refute (prove wrong) a claim.
Strategy: Find basalt on the *Mineral Composition of Igneous Rocks* chart (p. 14). Name two minerals from the chart as your evidence, and explain how they show the claim is wrong.

Base your answer to question 5 on the information below and on your knowledge of Earth and Space Sciences. Some questions may require the use of the *Reference Tables*.

Basalt of the Canary Islands

The Canary Islands are a chain of volcanic islands located in the Atlantic Ocean near the west coast of Africa. The islands were built mainly by basaltic lava flows over millions of years. Gran Canaria is one of the largest islands.

A student studying the mineral composition of a basalt sample from Gran Canaria made the following claim:

"The Gran Canaria basalt sample has a felsic composition and contains sodium-rich minerals."

5 Cite evidence about the composition of basalt to *refute* the student's claim. Justify your response by including at least **two** minerals as evidence. [1]

Base your answers to questions 6 through 8 on the information below and on your knowledge of Earth and Space Sciences. Some questions may require the use of the *Reference Tables*.

Invisible Gas at Mammoth Mountain

Mammoth Mountain is a volcano in eastern California, just south of Yosemite National Park. Magma deep beneath the mountain contains dissolved carbon dioxide gas. Since about 1990, carbon dioxide has been leaking out of the magma and seeping up through cracks and soil. In several areas around the mountain, the gas has killed large groups of trees by building up in the soil around their roots.

Visitors are warned that carbon dioxide can collect in low places, such as hollows in the ground, snow pits, and closed tents or cabins. The table below shows the approximate density of two gases at room temperature.

Gas	Approximate Density (g/L)
Air	1.2
Carbon dioxide	1.8

Skill: Use data to explain a cause and effect.
Strategy: Compare the two densities in the table. Ask what a denser gas does when it is surrounded by a less dense gas.

6 Based on the data table, why can carbon dioxide build up in low places like snow pits near Mammoth Mountain?

(1) Carbon dioxide is denser than air, so it sinks and collects in low places.

(2) Carbon dioxide is less dense than air, so it rises and gets trapped under snow.

(3) Carbon dioxide and air have the same density, so they mix evenly.

(4) Carbon dioxide is denser than air, so it rises quickly into the atmosphere.

Skill: Track matter moving between Earth systems.
Strategy: Find where the carbon starts in the reading and where it ends up. Name both systems (geosphere, hydrosphere, atmosphere, biosphere), then describe how it moves.

7 Identify the Earth system where the carbon dioxide at Mammoth Mountain comes from and the Earth system it moves into. Then describe how the carbon dioxide moves between them. [1]

Comes from: _____

Moves into: _____

How it moves: _____

_____

Some of the carbon in magma can come from carbonate rocks, such as limestone, that formed on the ocean floor and were later carried deep into Earth. The steps below, labeled A through D, describe how this carbon can return to the atmosphere.

A	Magma rises toward the surface beneath a volcano.
B	Limestone on the ocean floor is buried and carried deep into Earth.
C	Carbon dioxide leaks out of the magma and escapes into the air.
D	Rock in the mantle partially melts, forming magma that contains carbon.

Skill: Sequence the steps of a process.
Strategy: Find the step that has to happen first: where is the carbon at the start? Then put each step in order so that each one causes the next.

8 Which sequence correctly shows how carbon in limestone on the ocean floor can return to the atmosphere?

(1) A → B → C → D

(2) B → D → A → C

(3) D → A → B → C

(4) B → A → D → C

Skill: Read two graphs together.
Strategy: Find each location's elevation in the table. Then use that elevation on each graph to read the boiling point and the extra time needed. Higher elevation means lower air pressure.

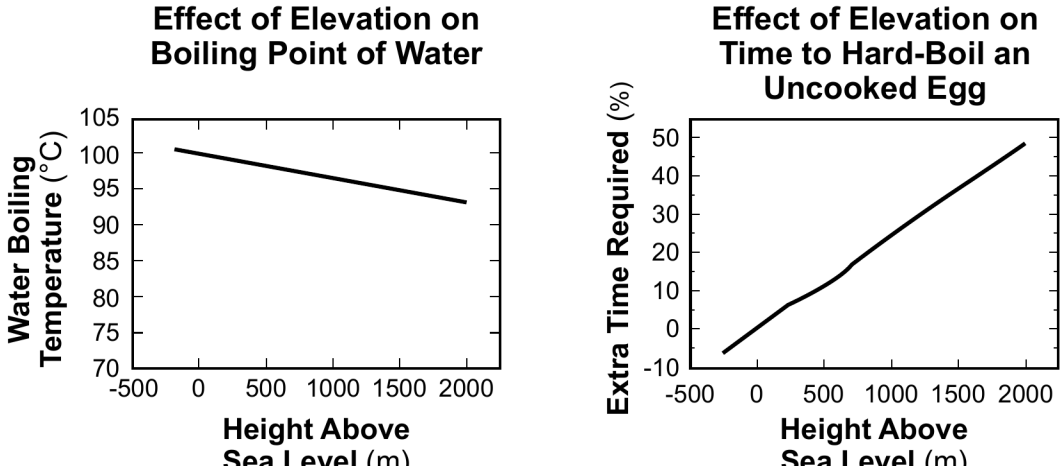
Base your answer to question 9 on the information below and on your knowledge of Earth and Space Sciences. Some questions may require the use of the *Reference Tables*.

Boiling Water at Different Elevations

A student practiced hard-boiling eggs at home in Caroga Lake, New York, before a cooking contest at the New York State Fair near Onondaga Lake. The elevation of a location can affect the time needed to hard-boil an egg.

Location	Elevation Above Sea Level (m)
Caroga Lake	450
Onondaga Lake	110

The graphs below represent some information about the temperature of boiling water and extra time required to hard-boil an uncooked egg at different elevations.



9 Which row contains the terms that correctly complete the statement below?

The boiling point of water at Caroga Lake is ___**1**___ than the boiling point of water at the location of the New York State Fair near Onondaga Lake. In order to transfer enough energy to the egg at Caroga Lake, ___**2**___ time is needed to ___**3**___ the temperature of the egg.

Row	1	2	3
(1)	greater	more	increase
(2)	less	more	increase
(3)	greater	less	decrease
(4)	less	less	decrease

Skill: Interpret a model (cause and effect).
Strategy: Trace the arrows from the olivine. Ask two questions: what happens to the carbon dioxide, and what happens to the pH? Pick the choice that matches both.

Base your answer to question 10 on the information below and on your knowledge of Earth and Space Sciences. Some questions may require the use of the *Reference Tables*.

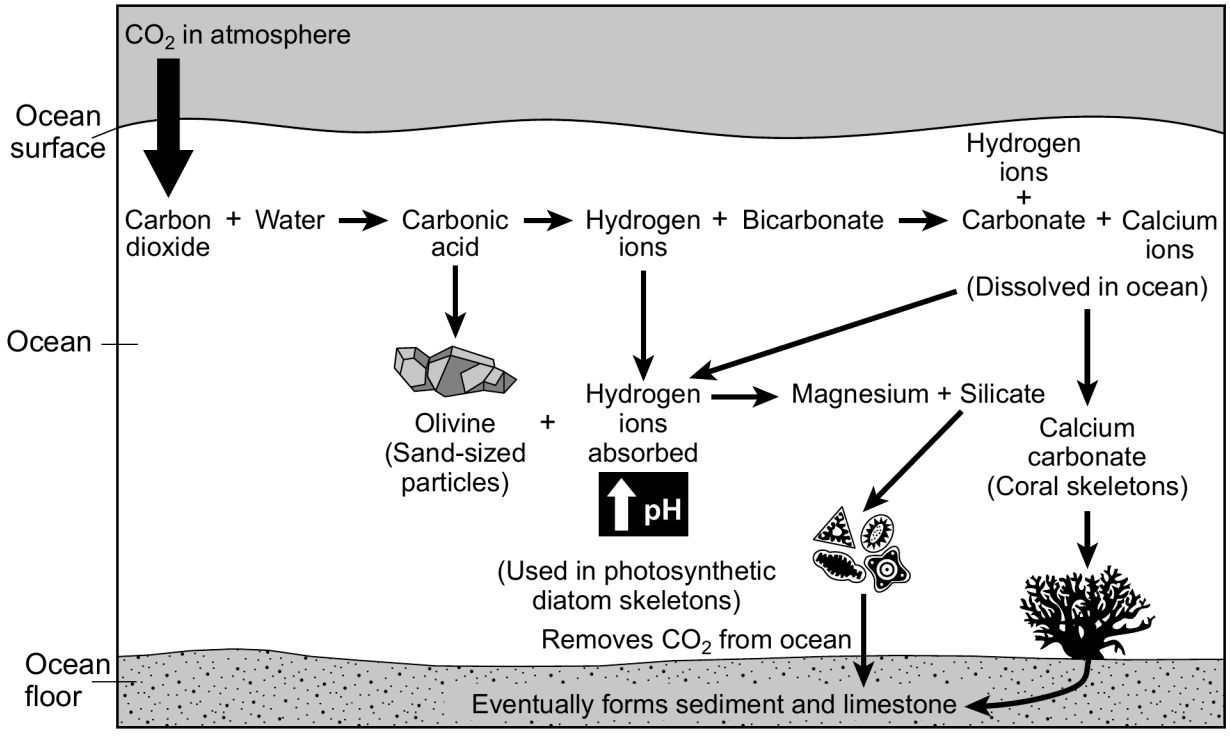
Coastal Carbon Capture

Mineral weathering, which takes place over geologic time scales, is one of the main mechanisms Earth uses to recycle carbon dioxide. In a process called Coastal Carbon Capture, a common volcanic mineral, olivine, is used to limit the impacts of CO_2 on the hydrosphere. In this process, olivine is mined from Earth, ground into sand-sized particles, and then transported to coastal regions where it is spread along shorelines. Once in the water, the olivine sand will absorb hydrogen ions, gradually making the water less acidic and potentially protecting wildlife like shellfish, corals, and fish.

In July 2022, this technique was performed at the eastern end of Long Island, New York, as part of an ongoing research project.

The model shows some information about the coastal carbon capture process.

Coastal Carbon Capture: Adding Olivine Sand to Oceans



10 Which claim correctly summarizes how the Coastal Carbon Capture process is a solution that will reduce the impact of ocean acidification?

(1) Olivine will lower the pH of the ocean and decrease CO_2 levels.

(2) Olivine will increase the pH of the ocean and decrease ocean CO_2 levels.

(3) Olivine will lower the pH of the ocean and increase ocean CO_2 levels.

(4) Olivine will increase the pH of the ocean and increase ocean CO_2 levels.

Skill: Describe a process using evidence from a reading.
Strategy: Underline what the water does at each step in the reading. Your answer needs both what happens to the calcite in the limestone and what happens when the water drips into the cave.

Base your answers to questions 11 through 13 on the information below and on your knowledge of Earth and Space Sciences. Some questions may require the use of the *Reference Tables*.

Minerals in Two National Parks

Carlsbad Caverns National Park in New Mexico has some of the largest caves in the world. The caves formed in thick layers of limestone, which is made mostly of the mineral calcite. Groundwater that is slightly acidic seeps down through cracks and dissolves some of the calcite. When this water drips from a cave ceiling, some carbon dioxide escapes from the water into the cave air, and calcite crystals form again. Over thousands of years, the crystals build long stalactites that hang from the ceiling.

11 Describe how water moves calcite to form a stalactite. Include what happens to the calcite in the limestone and what happens when the water drips into the cave. [1]

Students studied two minerals. Mineral 1 came from a cave formation in Carlsbad Caverns. Mineral 2 came from El Capitan, a granite cliff in Yosemite National Park. The students recorded their observations of each mineral.

Observations of Mineral 1 (Carlsbad Caverns)

A. The mineral is white to colorless with a vitreous (glassy) luster.
B. The sample was found 230 meters below the surface.
C. The mineral breaks along three cleavage directions that are not at 90°.
D. The mineral bubbles when a drop of acid is placed directly on it.
E. The sample has a mass of 45 grams.
F. A copper penny scratches the mineral, and the mineral does not scratch glass.

Skill: Select the evidence that matters.
Strategy: Cross out observations that can't identify a mineral, like where it was found or how big it is. Use the rest to follow the *Mineral Identification Flowchart* (pp. 16–17).

12 Based on the students' observations, which set of observations and mineral name correctly identifies Mineral 1?

(1) Observations A, B, and E identify calcite.

(2) Observations C, D, and F identify calcite.

(3) Observations C, D, and F identify halite.

(4) Observations A, B, and D identify gypsum.

Observations of Mineral 2 (El Capitan, Yosemite)

- Pink in color with a vitreous (glassy) luster
- Scratches a glass plate
- Breaks along two smooth cleavage planes that meet at about 90°
- No striations or dissolution lines on the cleavage surfaces

Skill: Use two reference tables together.
Strategy: Identify the mineral with the flowchart first. Then find it on the *Mineral Composition of Igneous Rocks* chart (p. 14) to decide felsic or mafic.

13 Use the *Mineral Identification Flowchart* to identify Mineral 2. Then use the *Mineral Composition of Igneous Rocks* chart to identify whether granite is felsic or mafic, and name **one** other mineral that is likely found in the El Capitan granite. [1]

Mineral 2: _____

Felsic or mafic: _____

Other mineral in granite: _____